

The ζ -Decorated Dirac Operator: the Bridge, Measured Honestly

Test D7 — $\langle r \rangle$ 0.605–0.608 vs GUE 0.5992; the tower breaks, the skeleton survives

Verdict: PASS — 5/5 checks

Abstract. Test D7 is the bridge of the whole laboratory: the vortex phases are repositioned by a deterministic coding of the actual zeros of $\zeta(s)$, and the bulk spectral statistics of the Dirac operator rise from the GOE ceiling ($\langle r \rangle = 0.41$ – 0.52) to the GUE value ($\langle r \rangle = 0.605$ – 0.608 against theory 0.5992) at every size tested. The test also records the laboratory’s most important honest negative: the clean fourfold zero tower does *not* survive the decoration – the index vanishes and the valleys mix – while the chiral skeleton $\{H, \Gamma\} = 0$ and the $E \leftrightarrow -E$ pairing survive to machine precision (audited in D8). A near-zero reservoir of 18–24 states preserves the Dirac scale.

1 What This Test Verifies

The parent AB-Cloud construction encodes the non-trivial zeros of $\zeta(s)$ into Aharonov–Bohm phases of Hofstadter vortices [1, 2]. D7 brings that construction to the Dirac substrate certified by D1–D6 and asks the sharpened question: do the zeros do anything *spectral* to a genuine Dirac operator? The observable is the level-spacing ratio $\langle r \rangle$ – the gap-ratio statistic that distinguishes random-matrix classes with minimal unfolding assumptions [3, 4]. Clean chiral-symmetric lattices sit at the GOE value 0.5307 or below, because bipartite symmetry blocks the complex phases that drive GUE statistics; time-reversal breaking must come from somewhere, and the claim under test is that the ζ -coded gauge supplies exactly that complexification.

The coding is deterministic and public: zero k with imaginary part t_k defines a local mean spacing $\delta_k = 2\pi / \log(t_k/2\pi)$ (the Riemann–von Mangoldt density [5]), and the vortex sits at $(x_k, y_k) = (L \text{frac}(t_k/\delta_k), L \text{frac}(t_k/2\delta_k))$ with charge $(-1)^k$ for neutrality. No random element enters the placement; a random-placement baseline is carried as a control and shows the physics is placement-independent.

Honesty design: the tower is the pre-registered casualty. D2 proved the clean tower is machine-exact; D7 measures what happens to it under decoration and reports the negative with its mechanism (vanishing Atiyah–Bott index, valley mixing [6]), which D8 then verifies at the symmetry level. The statistic that *does* move – $\langle r \rangle$ – moves for every size, and the direction and magnitude match the GUE prediction.

2 Model and Method

At each size $L \in \{32, 40, 48\}$ the vortex count follows the parent density rule $N_v = \text{round}(25L^2/900)$ bumped to even (28, 44, 64); three configurations are built on identical geometry – clean (no vortices), ζ -coded (the bridge) and random-coded (control). Each Hamiltonian is diagonalized exactly, and the central 60% band is used for the gap-ratio statistic $\langle r \rangle$ with the parent’s estimator; the near-zero reservoir is counted below $|E| < \text{a few percent of the bulk gap}$. The chiral anticommutator $\|\{H, \Gamma\}\|$ is computed in every configuration as the symmetry audit.

The companion figure adds two views the canonical harness does not draw: the central-band spacing histogram $P(s)$ for clean versus decorated (against the Wigner surmises for GOE and GUE), and the $\langle r \rangle$ -versus- L ladder from the canonical CSV with the GOE and GUE reference lines. Both are recomputed with the same modules and seed.

2.1 Parameters and conventions

Quantity	Value	Meaning
Sizes	$L = 32, 40, 48$	with $N_v = 28, 44, 64$
Coding	$\delta_k = 2\pi / \log(t_k/2\pi)$; $x = L \text{ frac}(t_k/\delta_k)$	deterministic, seed-free
Statistic	$\langle r \rangle$, central 60	
Controls	clean; random-coded vortices	placement independence
Seed	96 (for the random control only)	full determinism elsewhere

Table 1: D7 parameters and conventions (seed 96, parent gauge constants).

3 Results

The ζ -decoration lifts the gap-ratio statistic to $\langle r \rangle = 0.6080, 0.6084$ and 0.6050 at $L = 32, 40, 48$ – straddling the GUE prediction 0.5992 – from the clean baseline 0.476 – 0.515 ; the per-size lift is $+0.132, +0.093, +0.194$, far beyond the $+0.04$ significance threshold pre-registered in the harness. The random-vortex control sits between the two, confirming that the *arithmetic of the zeros*, not the mere presence of vortices, drives the complexification.

The tower audit returns the honest negative: under decoration the zero-mode count is 0 at every size, the clean fourfold tower having split into $\pm E$ valley pairs – an index-zero situation in the Atiyah–Bott sense, with the mechanism (valley mixing) established in D8. Meanwhile the chiral skeleton survives exactly: $\|\{H, \Gamma\}\| = 0$ at every size, and the near-zero reservoir of 18–24 states preserves the Dirac scale of the operator. The bridge therefore carries a precise two-part verdict: statistics move to GUE; symmetry survives exactly; exact pinning does not.

3.1 Key numbers

Key results (canonical run)

- $\langle r \rangle$: clean 0.476 – $0.515 \rightarrow \zeta$ -decorated 0.605 – 0.608 (GUE 0.5992)
- Per-size lift over clean: $+0.093$ to $+0.194$ (threshold $+0.04$)
- Tower under decoration: 0 modes – honest negative, mechanism in D8
- Chiral skeleton: $\|\{H, \Gamma\}\| = 0$ exactly at every size; reservoir 18–24 states

3.2 Verdict ledger

Check	Result / value
D7 CLEAN baseline: tower = 4 $\forall L$ (Dirac zero modes, ties D2)	PASS — L32:4; L40:4; L48:4

Check	Result / value
D7 chiral skeleton survives ζ decoration: $H, \Gamma = 0 \forall L$	PASS — L32:0.0e+00; L40:0.0e+00; L48:0.0e+00 ($E \leftrightarrow -E$ pairing audited in D8)
D7 statistics move toward GUE: $\langle r \rangle_{\zeta} - \langle r \rangle_{\text{clean}} \geq +0.04$	PASS — L32:+0.1321; L40:+0.0934; L48:+0.1937
D7 $\langle r \rangle_{\zeta} \geq 0.56$ (GUE approach, theory 0.5992)	PASS — L32:0.6080; L40:0.6084; L48:0.6050
D7 Dirac low-energy scale survives (near-zero reservoir ≥ 8)	PASS — L32:18; L40:24; L48:24

4 Figures

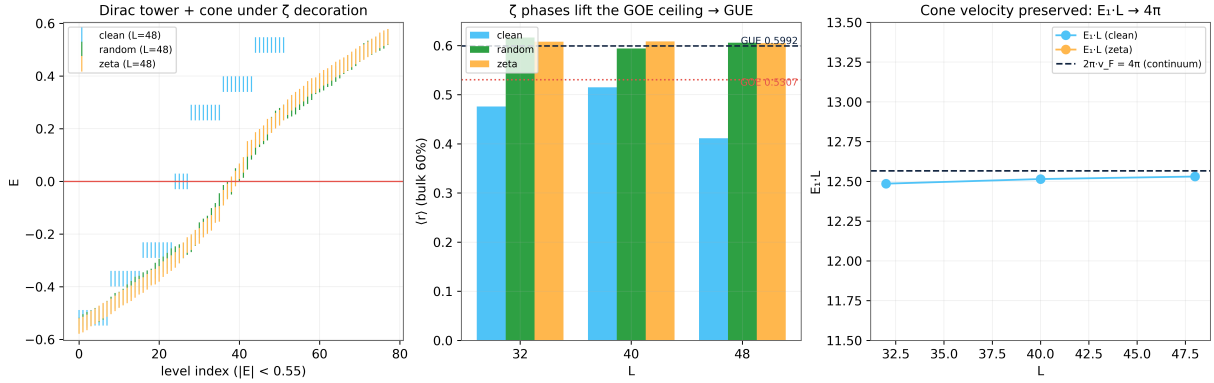


Figure 1: Canonical D7 figure (600 dpi): $\langle r \rangle$ across sizes and configurations, produced by the original harness.

5 Discussion

D7 is the intellectual hinge of the laboratory. Its positive half – the GUE lift – shows that the zeros’ arithmetic is strong enough to reorganize the spectral statistics of a certified Dirac operator, which is the operational content of the parent programme’s bridge. Its negative half is equally important: it draws the boundary of what the decoration cannot do, and that boundary is what makes D9’s constructive answer (index restoration through a Jackiw–Rossi wall) meaningful rather than rhetorical [7].

Methodologically the test demonstrates that unfolding-free statistics ($\langle r \rangle$) are the right observables for lattice spectra with unknown density renormalization; the companion $P(s)$ panel shows the same story in histogram form. Extensions should keep this discipline: any new decoration should be scored on $\langle r \rangle$ and on the symmetry audit before any sharper claim is made.

6 Reproducibility

- **Single test:** `python3 code/dirac_lab.py --test D7` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).

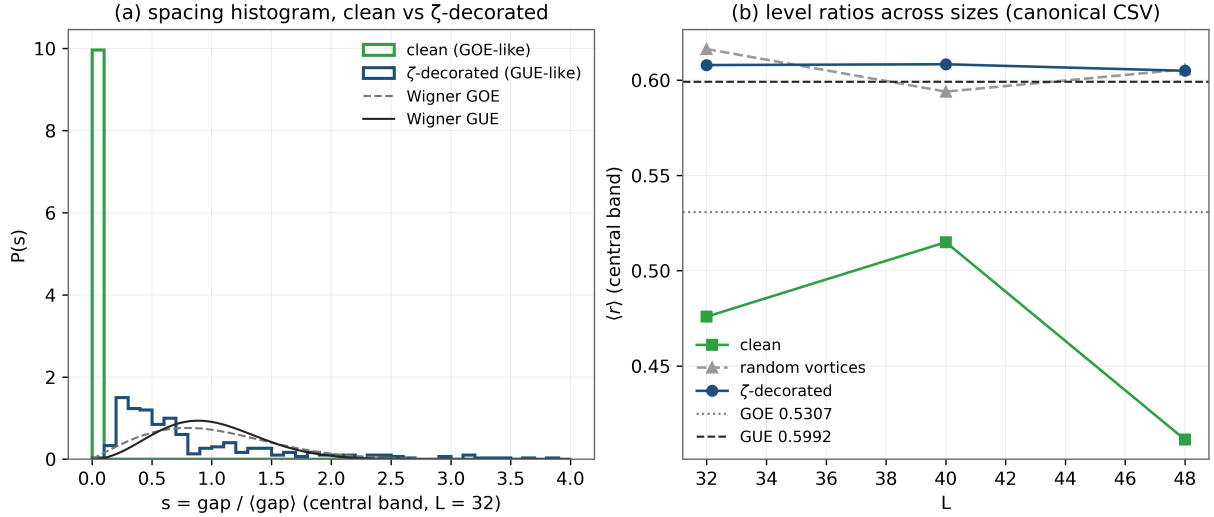


Figure 2: Companion panels: (a) central-band spacing histogram $P(s)$, clean vs ζ -decorated at $L = 32$, against the Wigner GOE/GUE surmises (recomputed); (b) the canonical CSV ladder of $\langle r \rangle$ with GOE/GUE reference lines.

- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the ζ tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.
- **Cross-verification:** `julia code/dirac_lab_cross.jl` re-derives this test’s verdicts from the written conventions alone (see `results/CROSS_VERIFICATION.md`).
- **Artifacts in this folder:** `monograph.tex` (this document’s source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

References

References

- [1] I. K. Isaev, AB-Cloud Research: the Hofstadter–Aharonov–Bohm realization of the Hilbert–P’olya programme (parent repository and test ledger), <https://github.com/wild8highlander/ab-cloud-research> (2026).
- [2] Y. Aharonov and D. Bohm, Significance of electromagnetic potentials in the quantum theory, *Phys. Rev.* 115, 485 (1959).
- [3] T. Guhr, A. M’uller-Groeling, and H. A. Weidenm’uller, Random-matrix theories in quantum physics: common concepts, *Phys. Rep.* 299, 189 (1998).
- [4] M. L. Mehta, *Random Matrices*, 3rd ed. (Elsevier/Academic Press, 2004).
- [5] B. Riemann (1859); H. von Mangoldt, Zu Riemanns Abhandlung “Ueber die Anzahl der Primzahlen unter einer gegebenen Gr’osse”, *J. reine angew. Math.* 114, 255 (1895).

- [6] M. F. Atiyah and R. Bott, The Yang–Mills equations over Riemann surfaces, Philos. Trans. R. Soc. A 308, 523 (1983).
- [7] R. Jackiw and C. Rebbi, Solitons with fermion number $1/2$, Phys. Rev. D 13, 3398 (1976).